02-620: Machine Learning for Scientists

Spring 2025

1. Course Information

1.1 Vital information

- Course
  - Lecture
    - Time: M/W/F 5:00PM-5:50PM EST
    - Location: Hamerschlag Hall (HH) B103
- Recitations
  - Section A
    - Time: Thursday 7:00PM-7:50PM EST
    - Location: Gates and Hillman Centers (GHC), 4215
  - Section B
    - Time: Friday 6:00PM-6:50PM EST
    - Location: Gates and Hillman Centers (GHC), 4215
- Instructors
  - **Min Xu**
    - Email: mxu1@cs.cmu.edu
    - Website: cs.cmu.edu/~mxu1
  - **Martin Zhang**
    - Email: martinzh@andrew.cmu.edu
    - Website: mzhanglab.github.io
- TAs
  - **Shiyi Du**
    - Email: shiyid@andrew.cmu.edu
  - **Minhyek Jeon**
    - Email: minhyekj@andrew.cmu.edu
  - **Yuxin Lu**
    - Email: yuxinlu@andrew.cmu.edu

1.2 Course Description

With advances in scientific instruments and high-throughput technology, scientific discoveries are increasingly made by analyzing large-scale data generated from experiments or collected from observational studies. Machine learning methods that have been widely used to extract complex patterns from large speech, text, and image data are now being routinely applied to answer scientific questions. The course will select scientific questions that arise in genomics, genetics, and medicine and discuss how to address these questions using machine learning techniques. It will cover disease-related genetic variant discovery with regression methods; clinical decision-making for patients with classification methods; pathway discovery with clustering algorithms; learning gene regulatory networks with probabilistic graphical models; genome sequence analysis with hidden Markov models; making

functional annotations of genomes with deep learning methods; and selecting an appropriate machine learning technique for the given scientific problem using learning theories. This course is intended for graduate students interested in learning machine-learning methods for scientific data analysis and modeling. Programming skills and basic knowledge of linear algebra, probability, and statistics are assumed. At the completion of the course, you are expected to be able to write basic machine-learning code from scratch. Homework assignments will consist of written problems and analyses of genetic and genomic data drawn from the literature in biology. The course grade will be computed as the result of homework assignments, midterm tests, project, and class participation.

1.3 Pre-requisites

Students should have programming skills, as well as basic knowledge of linear algebra, probability, and statistics. Have taken both 02-601 and 02-680 (or equivalent classes). Have basic familiarity with Python and Numpy programming. Be able to do simple linear algebra using Numpy and save/access Jupyter Notebooks.

1.4 Course Details

Piazza Homepage.

The course homepage will be hosted on Piazza. Piazza will be used as a central repository for course information. You should be automatically enrolled at Piazza.

Discussion Forum.

An online forum is provided on Piazza as an area for discussion and questions. The forum will be moderated by the course staff who will respond to questions, but students are encouraged to help each other via discussion. However, assignment specifics should not be discussed — any hints will be provided by the teaching staff. You can find the link to the class on Piazza.

Grades & Homework Submissions.

Gradescope will be the main place for homework submission and regrade requests. You can find the Gradescope page listed on the Piazza.

1.5 Textbooks

Reading material will be drawn from the following textbooks. All textbooks listed below are freely available online as pdf files.

Machine Learning: a Probabilistic Perspective by Kevin Murphy [M]

The Elements of Statistical Learning: Data Mining, Inference, and Prediction by Trevor Hastie, Robert Tibshirani, and Jerome Friedman [H]

Additional textbook:

Machine Learning by Tom Mitchell

2. Curriculum

2.1 Key dates

See the Calendar on Piazza for more information

2.2 Tentative course schedule

As with course topics, the lecture-by-lecture below is tentative and subject to change.

COURSE SCHEDULE

Week	Topic	Lecture	Reading
Week 1	Introduction to ML	Supervised/unsupervised learning MLE, MAP estimation	M Ch 1.1-1.3, 2.2.3, 3.1-3.4, 4.6.2.2
Week 2	Regression	Linear regression, regularization [*Finding genetic variants that control gene expression levels*]	M Ch 7.1-7.3.1, 7.5.1, 7.5.4, 13.3.1, 13.3.3, 13.3.4
Week 3	Classification	K-nearest neighbor method Naive Bayes classifier [*Genetic variants that affect disease state*] Logistic regression	H Ch 13.3; M Ch 1.4.2, M 3.5.1-3.5.3, 8.1-8.3, 8.6 M Ch 8
Week 4	Classification	Generative and discriminative classifier Decision tree, random forest [*Cancer classification based on expression*]	M Ch 16.1-16.2
Week 5	Clustering	Hierarchical clustering, K-means [*Gene module/pathway discovery*]	M Ch 25.1, 25.5; H Ch 13.1-13.2
Week 6	Clustering	SVM [*Cancer classification based on expression*]	M Ch 14.5
Week 7	Probabilistic graphical models Midterm I	Intro to Probabilistic graphical models Bayesian networks: models, learning, and inference [*Learning gene networks*]	M Ch 10
Week 8		Spring Break, No Classes	
Week 9	Latent variable models	Dynamic Bayesian networks, Gaussian graphical models [*Modeling longitudinal data*]	H Ch 17.1-17.3
Week 10	Latent variable models Dimensionality reduction	Mixture models, EM algorithm [*Gene module/pathway discovery with uncertainty*] PCA [*Population structure discovery in genome sequences*]	M Ch11.4.1-11.4.2, 11.4.7, 11.5 M Ch 11.2.1-11.2.3
Week 11	Latent variable models	Hidden Markov model, forward-backward algorithm, Baum-Welch algorithm	M Ch17.1.1-17.2.2, 17.3-17.5

Week 12	Bayesian learning Neural networks	Bayesian estimation, MCMC [*Modeling uncertainty in gene networks*] Models, backpropagation algorithm [*Predicting sequence specificity in functional genomics*]	H Ch 8.6; M Ch 15.1-15.2 M Ch 16.5
Week 13	Model selection Midterm II	Bias-variance trade-off [*Which model is the best?*]	H Ch 7.1-7.10
Week 14	Guest lectures	TBD	
Week 15		Project presentations	

3. Coursework

Coursework will consist of the following components. **You will have two late days in total that you can use to receive full credits for any homework submissions during the semester. For each additional late day used, 15% of the received grade will be deducted.**

Homework assignments. (50% of grade) Written homework assignments will test your knowledge of the material covered in class. See key dates in Piazza for homework due dates. Each homework will consist of written and programming assignments and will be posted on Piazza at least one week before the due date. Please use **Python** as the programming language to complete all homework assignments.

Examinations. (25% of grade) The exams will test your knowledge of the material from the class. The two midterms will be held in class.

- Midterm 1 (12.5% of grade): in-class expectation, date see Piazza
- Midterm 2 (12.5% of grade): in-class expectation, date see Piazza

The midterms will not be cumulative: midterm 2 will cover material encountered after midterm 1. You are allowed to bring one-page hand-written letter-sized note to each exam.

Project. (25% of grade) In the course project, you will analyze a dataset with methods of your choice as a team of 2-4 students. Details will be released

Attendance. (2% of grade, bonus)

More information will become available on suggested projects, detailed instructions on the format of the proposal, presentation, and final report.

4. Collaboration Policy and Academic Integrity

All classwork should be done independently unless explicitly indicated on the assignment handout. You may *discuss* homework problems with classmates, but must write your solution by yourself. Sharing written code or solutions is strictly prohibited. If you do discuss assignments with other classmates, you must supply their names at the top of your homework. No excuses will be accepted for copying others' work, and violations will be dealt with harshly. (Getting a bad grade is much preferable to cheating.)

The university's policy on academic integrity can be found at the following link: http://www.cmu.edu/academic-integrity/. In part, it reads, "Unauthorized assistance refers to the use of sources of support that have not been specifically authorized in this policy statement or by the course instructor(s) in the completion of academic work to be graded. Such sources of support may include but are not limited to advice or help provided by another individual, published or unpublished written sources, and electronic sources." You should be familiar with the policy in its entirety. **The default penalty for any academic integrity violation is a failure of the course. In particular: the use of a previous semester's answer keys or online solutions for graded work is absolutely forbidden. Any use of such material will be dealt with as an academic integrity violation.**

Generative AI policy: To best support your own learning, you should complete all graded assignments in this course yourself. You may use generative AI tools like ChatGPT for consultation, but your code and solutions must be entirely your own. If you have any questions about using generative AI in this course, please email or talk to the instructors.

5. Other Policies

Excused absences: Students claiming an excused absence for an in-class exam must supply documentation (such as a doctor's note) justifying the absence. Absences for religious observances must be submitted by email to the instructor during the first two weeks of the semester. Note that job or internship interviews are not a justification for an excused absence.

Other: The following policies of 15-110 also apply to this class. This text is mostly quoted from the 15-110 website (with some modifications):

- **I must be out of town for a university-related event (e.g. member of a team). What should I do about my assignments?**

  If you have an official excuse we will make special arrangements for you to submit the assignment, please contact the instructors.

- **I am out of town attending a family/important event. How can I submit my assignments due for the week?**

  The assignment must be submitted online before the due date.

- **I missed the in-class exam because I fell sick. What should I do?** You must immediately seek medical treatment and receive an official medical excuse. You must also contact the instructors

prior to the exam or as soon as possible. If you can produce documentation we can make arrangements to give you a makeup test. Otherwise, we will be unable to make any exceptions.

- **I am failing the course. Is there any extra work I can do to get a passing grade?** Unfortunately, we cannot make exceptions. The best way to avoid this situation is to talk to one of the instructors as soon as possible to find out what you need to do. Do not wait until the last few weeks of classes to discuss your performance.

- Further exceptions can be granted only in case of medical emergencies, family/personal emergencies (death of a family member), and university-approved absences (e.g., traveling for a conference, etc.). In case of such emergencies, you need to contact appropriate authorities. They will inform the student affairs office about your issue and the student affairs office will inform us. Only in such a case, we have the ability to grant you further exemptions. The authorities include Student Affairs (Submitting CMU Cares Referral Form), UHS for medical emergencies, CaPS for personal emergencies.

6. Accommodations for Students with Disabilities

If you have a disability and have an accommodations letter from the Disability Resources office, we encourage you to discuss your accommodations and needs with us as early in the semester as possible. We will work with you to ensure that accommodations are provided as appropriate. If you suspect that you may have a disability and would benefit from accommodations but are not yet registered with the Office of Disability Resources, we encourage you to contact them at access@andrew.cmu.edu.

7. Provost's Statement on Student Well-Being

Take care of yourself. Do your best to maintain a healthy lifestyle this semester by eating well, exercising, avoiding drugs and alcohol, getting enough sleep, and taking some time to relax. This will help you achieve your goals and cope with stress.

All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful.

If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. Counseling and Psychological Services (CaPS) is here to help: call 412-268-2922 and visit their website at http://www.cmu.edu/ counseling/. Consider reaching out to a friend, faculty, or family member you trust for help getting connected to the support that can help.

If you or someone you know is feeling suicidal or in danger of self-harm, call someone immediately, day or night:

CaPS: 412-268-2922

Resolve Crisis Network: 888-796-8226

If the situation is life-threatening, call the police:

On campus: CMU Police: 412-268-2323.

Off campus: 911

If you have questions about this or your coursework, please let us know.

8. Diversity, Equity and Inclusion

Throughout the semester, we will highlight the contribution of diverse groups of scientists in computer science, statistics, and mathematics. As we discuss applications of these mathematical concepts and techniques to problems in biology, we will draw materials from the literature based on the work of computational biologists with diverse backgrounds.

We are deeply committed to ensuring every student in our class feels supported and empowered. Remember we are here to help! We also encourage you to discuss these topics with the class staff. We are always open to suggestions from students on how to make the course more inclusive.

9. Student Academic Success Center (SASC)

SASC focuses on creating spaces for students to engage in their coursework and approach learning through a variety of group and individual options. We offer many opportunities for students to deepen their understanding of who they are as learners, communicators, and scholars. Our services and workshops are free to the CMU community and meet the needs of all disciplines and levels of study. SASC programs to support student learning include the following (program titles link to webpages):

- **Academic Coaching**--This program provides holistic, one-on-one peer support and group workshops to help undergraduate and graduate students implement habits for success. Academic Coaching assists students with time management, productive learning and study habits, organization, stress management, and other skills. Request an initial consultation here.
- **Peer Tutoring**--Peer Tutoring is offered in two formats for students seeking support related to their coursework. Drop-In tutoring targets our highest demand courses through regularly scheduled open tutoring sessions during the fall and spring semesters. Tutoring by appointment consists of ongoing individualized and small group sessions.You can utilize tutoring to discuss course related content, clarify and ask questions, and work through practice problems. Visit the webpage to see courses currently being supported by Peer Tutoring.
- **Communication Support**--Communication Support offers free one-on-one communication consulting as well as group workshops to support strong written, oral, and visual communication in texts including IMRaD and thesis-driven essays, data-driven reports, oral presentations, posters and visual design, advanced research, application materials, grant proposals, business and public policy documents, data visualisation, and team projects. Appointments are available to undergraduate and graduate students from any discipline at CMU. Schedule an appointment (in-person or video), attend a workshop, or consult handouts or videos to strengthen communication skills. Specific resources for multilingual students are also available.
- **Language and Cross-Cultural Support**--This program supports students seeking help with language and cross-cultural skills for academic and professional success through individual and group sessions. Students can get assistance with writing academic emails, learning expectations and strategies for clear academic writing, pronunciation, grammar, fluency, and more. Make an appointment with a Language Development Specialist to get individualized coaching.
- **Supplemental Instruction (SI)**--This program offers a non-remedial approach to learning in historically difficult courses at CMU. It utilizes a peer-led group study approach to help students succeed and is facilitated by an SI leader, a CMU student who has successfully completed the course.

SI offers a way to connect with other students studying the same course, a guaranteed weekly study time that reinforces learning and retention of information, as well as a place to learn and integrate study tools and exam techniques specific to a course. Visit the website to see courses with SI available here.